

WHEELS OF CHANGE
A farmer in Amritsar using the happy seeder in his fields

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SOLUTIONS

- **The Punjab govt is providing equipment to replough stubble while seeding the rabi crop**
- **Subsidised machinery is being made available through Agricultural Mechanisation for In-Situ Management of Crop Residue scheme in Punjab, Haryana, UP, Delhi**
- **Centre, Punjab are working together to promote non-paddy crops**

sow the next rabi crop, forcing farmers to burn the stubble to clear the fields. Data from the Indian Agricultural Research Institute's satellite monitoring division suggest that of the 35,350 crop residue-burning events in 2023, 65 per cent were from Punjab. Per estimates, 10 million tonnes of stubble is still burned every year.

Cognizant of the problem, the government has been trying to incentivise farmers away from stubble burning. For one, the central government has been financing the Promotion of Agricultural Mechanisation for In-Situ Management (PAMIM) of crop residue scheme in Punjab, Haryana, Uttar Pradesh and Delhi from FY19 to subsidise farm equipment. Machinery such as happy seeders and super seeders not only pull out the stubs and convert them into bales but also help with better seeding. The bales are carted to power plants, where they can be used to generate biomass fuel. Per agriculture ministry data, nearly 120,000 seeder machines were made available to farmers in Punjab this year and another 76,000 to those in Haryana. The Centre is also pushing these states to set up pelletisation and torrefaction plants to process paddy straw and help farmers earn some income, but the plans remain in various stages of implementation. Together, these plants can process 1.4 million tonnes of straw. Farmers in Punjab produce 22 MT of stubble annually. On paper, Punjab has a target of managing 16.27 MT of stubble, including 11.5 MT through in-situ management and 3.47 MT through ex-situ management. However, even ex situ management will require all 23 districts to have 4-5 stubble-to-pellets or torrefaction plants. Even if these pellets reach the thermal power plants for 20 per cent blending, it will increase the cost of power generation by about 10 paise per unit.

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Farmers themselves have yet to embrace these measures with enthusiasm. Jaswinder Sidhu, a farmer working in his field in Ropar tells us why. The state government, he says, maybe providing equipment for reploughing the stubble while seeding the rabi crop, but for that he needs tractors with 75 horsepower. Most of the tractors in Punjab, however, are only 50 hp-strong and hiring or buying tractors just to pull out stubble is too costly for the state's small and marginal farmers. Not to mention the cost of diesel to run the machine. Farm unions have for the past six years been asking for additional money to clear their fields, and not burn stubble.

The Supreme Court bench hearing the petitions on Delhi's pollution called for measures to sensitise farmers to end the stubble burning pattern and switch to other crops. Paddy is a water-guzzling crop—the Indian Council of Agricultural Research (ICAR) estimates that it takes 2,500-5,500 litres of water to produce a kilo of rice. With the water table receding, Punjab's farmers have deployed electricity-intensive pumps to extract groundwater.

As for crop diversification, the experience has not been a happy one for Punjab's farmer so far. In the past two decades, the state first tried growing citrus fruits and then dabbled in cotton cultivation. Last year, Chief Minister Bhagwant S. Mann personally tried to convince farmers to grow pulses and assured them of procurement at an MSP (minimum support price). But the state did not have the infrastructure to buy pulses, forcing many Punjab farmers to return to paddy. The apex court has asked the state to look at the model neighbouring Haryana has employed to shift from paddy. The area under cultivation of this crop has gone down from 1.56 million hectare in 2019 to below 1.3 mha now. That's because the state offers income support of Rs 7,000 per acre to the farmers if they move away from paddy and grow another crop.

Ultimately, it's collective will—of the government, individuals, industry, builders and cultivators—that can deliver us from the evil that hangs in the air of our cities. Till that is exercised, the solutions will keep coming, but they will remain only on paper. ■